

SUMIT CHUGH

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SUMMARY

Electrical Engineering student at Penn State with hands-on experience in PCB and analog circuit design, hardware validation, embedded systems, and CMOS/VLSI.

EDUCATION

The Pennsylvania State University

B.S. Electrical Engineering, Dean's List

University Park, PA

Aug. 2023 – May 2027

Relevant Coursework: Electronic Circuit Design, Digital Integrated Circuits, Circuits & Devices, Digital Design, Computer Organization, Discrete-Time Linear Systems

TECHNICAL SKILLS

Hardware/EDA: PCB Design (KiCad, Fusion 360), Cadence Virtuoso, Vivado, Multisim, ESP32, FPGA, CAD

Test/Instrumentation: NI LabVIEW/DAQmx, myDAQ, Oscilloscopes, Function Generators, Multimeters, Hardware Validation, Serial Data Acquisition

Programming: C/C++, Verilog, MATLAB, Arduino, Python (NumPy, Pandas, scikit-learn), Git

EXPERIENCE

Undergraduate Researcher

E-BIO-E Lab

Penn State, University Park, PA

May 2026 – Present

- Designed and debugged an ESP32-controlled PCB with DAC bias control, op-amp gain stages, transimpedance amplification, ADC acquisition, and RC filtering for a precision analog sensing front-end.
- Researching neuromorphic transistor approaches for low-power analog signal processing and feature extraction from analog sensor readouts.
- Built Python tools for real-time acquisition, filtering, plotting, and comparative signal analysis.
- Iterated 5×5 hollow array geometry, tuning needle, lumen, side-hole, and substrate dimensions using CAD, printing, microscopy, and mechanical testing.

Math Grader

Undergraduate Mathematics Courses

The Pennsylvania State University

Aug. 2024 – Present

- Grade undergraduate calculus and algebra assignments with consistent, accurate evaluation.

PROJECTS

GaitGuard – Real-Time Fall Detection System

ESP32, Python, scikit-learn, C++, Flask

- Built an ESP32 wearable fall-detection system using accelerometer processing, baseline calibration, FSM logic, and secondary sensor confirmation; won Best Philanthropic Hack - Bloomberg at Bitcamp 2026.
- Combined rule-based detection with Logistic Regression and built a Python serial bridge for live parsing, inference, dashboard streaming, and push notifications.

Autonomous Multi-Sensor Rover

ESP32-S3, ESP32-CAM, C++, Sensors, Robotics, Wi-Fi

- Developed a dual-controller autonomous rover using an ESP32-S3 for motor and sensor control and an ESP32-CAM for live video, with autonomous obstacle avoidance and browser-based manual driving modes.
- Structured the embedded architecture around ultrasonic sensing, Wi-Fi control, real-time camera streaming, battery monitoring, data logging, and manual override/emergency-stop features, with support for GPS, IMU, and encoder-based navigation.
- Laid out a custom PCB using Fusion 360.

Tone Control & Karaoke Circuit

PCB, Op-Amps, Lab Instruments

- Characterized a multi-stage analog audio circuit, measuring gain and frequency response and tracing signal-path faults using oscilloscopes and bench instruments.

Traffic Signal FSM Controller

LabVIEW, DAQmx, CMOS Logic, Atmel MCU

- Implemented a six-state controller using discrete CMOS logic and an Atmel MCU; created LabVIEW/DAQmx routines to verify state transitions, timing, and fault conditions across 12+ input cases.

8-Bit Pipelined Carry Lookahead Adder

Cadence Virtuoso, CMOS, Digital IC Design

- Built a hierarchical 8-bit carry lookahead adder in Cadence Virtuoso by combining two 4-bit CLA blocks with reusable CMOS logic cells and propagate/generate carry logic.
- Integrated flip-flops for pipelined operation and verified sum, carry, and timing behavior through transient simulations, including the 11111111 + 00000001 carry-out case.

Five-Stage MIPS Pipelined CPU

Verilog, Vivado, FPGA

- Architected a 5-stage pipelined MIPS processor with forwarding and hazard-detection logic; verified functionality across 20+ test cases using waveform simulation and FPGA implementation.